Pavan Kumar

Senior Data Engineer

pavanrk4512@gmail.com | +1 (316)-928-0588

PROFESSIONAL SUMMARY

Highly skilled Senior Data Engineer with a demonstrated track record of successfully delivering end-to-end data engineering solutions. Equipped with overall **5+** years of experience in architecting and implementing scalable data pipelines and optimizing data infrastructure for improved performance and efficiency. Proficient in leveraging cloud-based technologies and big data tools to drive data-driven decision-making and innovation. Adept at collaborating with cross-functional teams to design and deploy robust data solutions that align with business objectives.

- **Data Engineering**: Proficient in designing and implementing robust data engineering solutions using technologies such as Apache Spark, Apache Kafka, Hadoop, and cloud-based data services. Solid understanding of data ingestion, data transformation, data modeling techniques and data visualization.
- **Data Architecture**: Extensive experience in designing and implementing scalable data architectures, including data lakes, data warehouses, and real-time streaming platforms. Expert in data modeling techniques and optimizing data structures for efficient querying and analytics.
- **ETL Development**: Expertise in building end-to-end Extract, Transform, Load (ETL) processes to integrate and transform data from various sources. AWS or custom Python-based solutions for data integration and processing.
- **Big Data Technologies**: Strong knowledge of distributed computing frameworks like Hadoop and Spark, along with relatedtools such as Hive, Pig, and HBase. Experienced in working with large-scale data sets and implementing efficient data processing workflows.
- **Programming and Scripting**: Proficient in programming languages like Python, R, Java, and Scala. Skilled in writing complexSQL queries for data manipulation and retrieval. Familiarity with scripting languages like Bash or PowerShell for automationand orchestration tasks.
- **Cloud Data Platforms**: Hands-on experience with cloud-based data platforms such as AWS, GCP, or Azure. Seasoned in leveraging cloud services like Amazon S3, Google Cloud Storage, or Azure Data Lake Store for data storage and processing.Experience with cloud-native data services like AWS Glue, GCP Dataflow, or Azure Databricks.
- **Data Quality and Governance**: Strong understanding of data quality and data governance principles. Competent in implementing data validation checks, data lineage tracking, and data cataloging. Familiarity with data privacy regulations and best practices for ensuring data security using Collibra and Alation.
- **Team Collaboration**: Excellent communication and interpersonal skills, with a proven ability to collaborate effectively withcross-functional teams and stakeholders using Agile methodology. Skilled in leading and mentoring junior data engineers, fostering a collaborative and productive work environment.

EDUCATION

- **Masters in Industrial Engineering from Wichita State University.**

TECHNICAL SKILLS

Languages	Python, R, SQL, Java, C#, Unix Shell Scripting, Scala
Databases	MySQL, SQL Server, Oracle, Postgres, Mongo DB, Neo4j, Cassandra, IBM Db2, Azure Databricks
Hadoop Ecosystem	Spark, Hadoop, HDFS, Hive, Kafka, Pyspark, Yarn
Data Engineering Tools **Data Warehouse Tools** **Data Governance**	Airflow, Dremio, Alation, Infogix, Looker, Talend Snowflake, Data Bricks, Big Query, Redshift, Db2 Collibra, Alation
Cloud services	AWS- EC2, EMR, S3, RDS, IAM, ECS, SQS, QuickSight Azure Maria DB, CloudWatch, CloudFormation, Azure Cosmo DB, Microsoft Azure Data Factory, Virtual machines, Blob storage, GCP
Versioning Tools	GIT, Bitbucket, GitHub
Operating systems	Linux/Unix, Ubuntu, Windows,
Scrum Methodologies	Agile methodology, Waterfall model

PROFESSIONAL EXPERIENCE

Senior Data Engineer **Dec 2022 to Present**

CNA Insurance, Chicago, IL

Responsibilities:

- Managed the development and execution of ETL data pipelines in Azure Data Factory, improving the processing of millions of auctions transactions daily and facilitating faster auctions reporting.
- Administered Azure Data Lake and Azure SQL Database for optimal data storage solutions, preserving customer data for over 10 million accounts and ensuring high availability.
- Leveraged Azure Databricks and HDInsight for Big Data processing, enhancing credit risk modelling and customer segmentation, and driving actionable insights for data-based decision making.
- Implemented robust data security measures using Azure Security Center and Azure Key Vault, protecting sensitive auctions information, and ensuring compliance with GDPR and other auctions industry regulations.
- Orchestrated automated loan approval workflows using Azure Logic Apps, reducing processing time and enhancing customer experience.
- Worked with Azure DevOps to streamline CI/CD pipelines, enabling faster updates to customer-facing banking applications and reducing time-to-market for software releases.
- Streamlined deployment and scalability of data processing environments using Docker containerization and Azure.
- Implemented real-time data processing using Spark Streaming and PySpark, enabling rapid insights from transaction data and improving fraud detection and risk management.
- Enhanced data quality and integrity by ingesting and transforming data from diverse sources, and developed ETL processes using SSIS and SSRS packages to facilitate data-driven decision-making.
- Played a pivotal role in cross-functional teams to develop data models and solutions that drove a 20% improvement in meeting business needs.
- Implemented robust data governance and security measures, safeguarding sensitive customer data, and ensuring compliance with banking industry regulations.
- Boosted system efficiency by optimizing data pipelines and queries, and improved speed of generating banking reports using efficient Hive queries.
- Enhanced data processing efficiency by working extensively with RDDs, Data frames, and Hive for data analysis and processing, and optimized batch processing of streaming data with Spark Streaming.

Tools & Technologies: Azure Databricks, Data Factory, Logic Apps, EventHub, Spark Streaming, Terraform, Azure DevOps, YAML, Oracle, HDFS, MapReduce, YARN, Spark, Hive, SQL, Python, Scala, PySpark, Kafka, Power BI.

Senior Data Engineer

Progressive Insurance, Overland Park, KS **Sep 2020 to Nov 2022**

Responsibilities:

- Oversaw end-to-end operations of ETL data pipelines in Azure Data Factory, maintaining high-quality data integration, transformation, and loading processes.
- Implemented Logic Apps to streamline order processing workflows and enhance operational efficiency in the insurance industry, resulting in improved order fulfilment and streamlined data exchange.
- Developed and implemented data ingestion pipelines using Azure Synapse Analytics to efficiently extract, transform, and load (ETL) large volumes of structured and unstructured data into the data warehouse.
- Designed and optimized data models and schemas within Azure Synapse Analytics to ensure efficient storage, retrieval, and querying of data, resulting in improved performance and reduced costs.
- Leveraged Snowflake to optimize data warehousing and analytics capabilities in the insurance industry, enabling valuable insights, data-driven decision-making, and improved targeted marketing campaigns.
- Deployed Azure Event Hub for real-time data ingestion in the insurance industry, facilitating real-time monitoring of foot traffic, enhanced fraud detection, and improved operational agility.
- Managed the orchestration of Terraform for infrastructure automation in Azure deployments, streamlining provisioning and management of resources.
- Utilized Ansible for configuration management, automating infrastructure provisioning and ensuring consistency across environments.

- Utilized Hive within the Azure ecosystem to derive valuable insights, optimize queries, enable efficient data warehousing, and deliver comprehensive insurance analytics as an Azure Data Engineer, enabling data-driven decision-making and enhancing business performance.
- Employed PySpark and Scala in conjunction with Azure HDInsight, Kafka, and Spark Streaming to enable advanced data processing, real-time streaming analytics, and scalable data solutions within the Azure ecosystem, driving actionable insights and enhancing decision-making capabilities in the insurance industry.
- Implemented and managed containerization using Docker, enhancing application scalability and portability.
- Developed and maintained Java-based applications using Spring Boot, ensuring high performance and reliability.
- Effectively utilized Oozie and Zookeeper for orchestration and coordination of data workflows in Azure, optimizing storage efficiency with ORC, Avro, Parquet, and delimited file formats as part of Azure data solutions, ensuring seamless data processing and efficient data storage in the insurance data engineering landscape.
- Collaborated with DevOps teams to orchestrate and manage containerized applications on Kubernetes, optimizing resource utilization.
- Ensured data security through encryption while seamlessly handling data ingestion, protecting sensitive data and maintaining data integrity in compliance with industry regulations in the insurance industry.
- Supervised the development and deployment of SSIS/SSRS packages for improved data accuracy and business insights in the insurance industry, enabling data-driven decision-making, generating comprehensive reports, and facilitating data visualization.
- Prepared technical specifications, data flow diagrams, and process documentation, fostering clear communication and effective collaboration within the team.

Tools & Technologies: Azure Databricks, Data Factory, Logic Apps, EventHub, Spark Streaming, Data pipeline, Terraform, Azure DevOps, Oracle, HDFS, MapReduce, YARN, Spark, Hive, SQL, Python, Scala, PySpark, Kafka, Power BI.

WellCare, Tampa, FL **April 2019 to Aug 2020**
Big Data Engineer

Responsibilities:

- Demonstrated hands-on experience in Azure Cloud Services, including Azure Synapse Analytics, SQL Azure, Data Factory, Azure Analysis Services, Application Insights, Azure Monitoring, Key Vault, and Azure Data Lake.
- Created batch and streaming pipelines in Azure Data Factory (ADF) using Linked Services, Datasets, and Pipelines to efficiently extract, transform, and load data.
- Developed Azure Data Factory (ADF) batch pipelines to ingest data from relational sources into Azure Data Lake Storage (ADLS Gen2) in an incremental fashion, applying necessary data cleansing and subsequently loading it into Delta tables.
- Implemented Azure Logic Apps to trigger automated processes upon receiving new emails with attachments, efficiently loading the files into Blob storage.
- Implemented CI/CD pipelines using Azure DevOps in the cloud, utilizing GIT, Maven, and Jenkins plugins for seamless code integration and deployment.
- Built a Spark Streaming application for real-time analytics on streaming data, leveraging Spark SQL to query and aggregate data in real-time and visualize the results in Power BI or Azure Data Studio.
- Developed Spark Streaming applications that integrate with event-driven architectures such as Azure Functions or Azure Logic Apps, processing events in real-time and triggering downstream workflows based on the results.
- Designed and implemented data pipelines using Lambda functions to ingest streaming data from various sources.
- Utilized AWS S3 for temporary storage of raw data and checkpointing and AWS Redshift for complex transformations and aggregations.
- Utilized AWS SQS for decouple data ingestion from processing for scalability and reliability.
- Defined and enforced data access controls based on user roles and permissions RBAC.
- Built ML models with cross functional teams using AWS Sage Maker to predict target variables using real-time and historical data.
- Created Hive tables, loaded and analyzed data using Hive queries, and developed custom Hive UDFs to extend Hive's functionality.
- Integrated data from various sources such as databases, APIs, logs, and files by writing Java code to extract, transform, and load (ETL) data into the desired format.
- Migrated ETL processes from Oracle to Hive, testing and validating the ease of data manipulation and processing in Hive.
- Implemented solutions to reprocess failure messages in Kafka using offset IDs, ensuring data integrity and reliability.
- Tuning queries to improve performance within PostgreSQL, minimizing unnecessary data retrieval and I/O operations.

- Understood the strengths and limitations of NoSQL databases like Cosmos DB, and structuring data efficiently for querying.
- Debugging data pipeline issues, optimizing performance within PostgreSQL and across the data ecosystem, and scaling infrastructure as needed.
- Developed a PySpark job in Scala to index data into Azure Functions from external Hive tables stored in HDFS.
- Utilized HiveQL to analyze partitioned and bucketed data, optimizing query performance and efficiency.
- Developed and executed Hive queries on analyzed data for aggregation and reporting purposes.
- Developed Sqoop jobs to efficiently load data from RDBMS into external systems like HDFS and Hive.
- Developed Spark applications using PySpark and Spark SQL for data extraction, transformation, and aggregation from multiple file formats, ensuring efficient data processing.
- Implemented Spark scripts using Scala and Spark SQL to access Hive tables for faster data processing.
- Loaded data from UNIX file systems to HDFS, ensuring data availability for further processing and analysis.
- Configured Spark Streaming to receive real-time data from Apache Flume and stored the stream data in Azure Tables using Scala.
- Utilized Spark RDD transformations to filter data and enhance Spark SQL processing capabilities.
- Utilized Hive Context and SQL Context to integrate Hive meta store and Spark SQL for optimized performance and data processing.
- Utilized GIT as the version control system to access repositories and coordinate with CI tools for effective code management and collaboration.

Environment: Azure Data Factory, Azure Synapse Analytics, Azure DevOps, AWS S3, AWS Redshift, AWS Glue, AWS Sage Maker, Sqoop, HDFS, Power BI, Git, Zookeeper, Flume, Kafka, Apache PySpark, Spark SQL, Scala, Hive, Hadoop, Cloudera, HBase, HiveQL, MySQL